Step 1. Install three tools

Install	From	Check
Git	https://git-scm.com	`git --version`
Docker Desktop	https://www.docker.com/products/docker-desktop	`docker --version`
OpenSSL	already on Mac and Linux	`openssl version`

Start Docker Desktop and wait until it says **running**. Then open Docker Desktop > Settings > Resources > Memory and set **8 GB**.

> Windows: first install WSL (PowerShell as administrator: `wsl --install`, restart), install Docker Desktop, switch on WSL integration for Ubuntu, and type every command in this guide in the **Ubuntu** window.

Step 2. Clone the project

```
TERMINAL
git clone https://github.com/F9info/kdl-os.git
cd kdl-os
```

Stay inside the `kdl-os` folder for every command that follows.

Step 3. Create the settings file (.env)

Paste the whole block into the terminal. It writes `.env` with fresh random secrets.

TERMINAL

```
PG_PASS=$(openssl rand -hex 16)
REDIS_PASS=$(openssl rand -hex 16)
MINIO_PASS=$(openssl rand -hex 16)
cat > .env <<EOF
NODE_ENV=development
APP_PORT=4000
FRONTEND_URL=http://localhost
BACKEND_URL=http://localhost:4000
AI_SERVICES_URL=http://localhost:5000
CORS_ORIGIN=http://localhost,http://localhost:3000,http://localhost:3001
JWT_SECRET=$(openssl rand -hex 32)
JWT_REFRESH_SECRET=$(openssl rand -hex 32)
JWT_EXPIRES_IN=15m
JWT_REFRESH_EXPIRES_IN=7d
APP_ENCRYPTION_KEY=$(openssl rand -hex 32)
POSTGRES_USER=postgres
POSTGRES_PASSWORD=$PG_PASS
POSTGRES_DB=kdl_db
DATABASE_URL=postgresql://postgres:$PG_PASS@localhost:5433/kdl_db
REDIS_PASSWORD=$REDIS_PASS
REDIS_URL=redis://:$REDIS_PASS@localhost:6380
MINIO_ROOT_USER=kdl_minio
MINIO_ROOT_PASSWORD=$MINIO_PASS
MINIO_ACCESS_KEY=kdl_minio
MINIO_SECRET_KEY=$MINIO_PASS
MINIO_ENDPOINT=localhost
MINIO_PORT=9000
MINIO_USE_SSL=false
MINIO_BUCKET=kdl-media
MEILISEARCH_HOST=http://localhost:7700
MEILI_MASTER_KEY=$(openssl rand -hex 24)
MEILISEARCH_API_KEY=replace-after-first-boot
MEILI_SEARCH_API_KEY=replace-after-first-boot
CHROMADB_URL=http://localhost:8000
ANTHROPIC_API_KEY=local
OPENROUTER_API_KEY=local
EOF
chmod 600 .env
```

Check with `ls -la .env` (about 2 KB). Never share or upload this file. No accounts or API keys are needed to run KDL OS. AI features need real keys later.

Step 4. Build and start everything

```
TERMINAL
docker compose up -d --build
docker compose ps
```

The first build takes **10 to 20 minutes**; later starts take seconds. Wait until `backend` shows **healthy** (about a minute). The database migrations run by themselves when the backend starts.

Service	Address
App through nginx (use this)	http://localhost
Frontend directly	http://localhost:3001
Backend API	http://localhost:4000
MinIO console (files)	http://localhost:9003
MeiliSearch (search)	http://localhost:7700

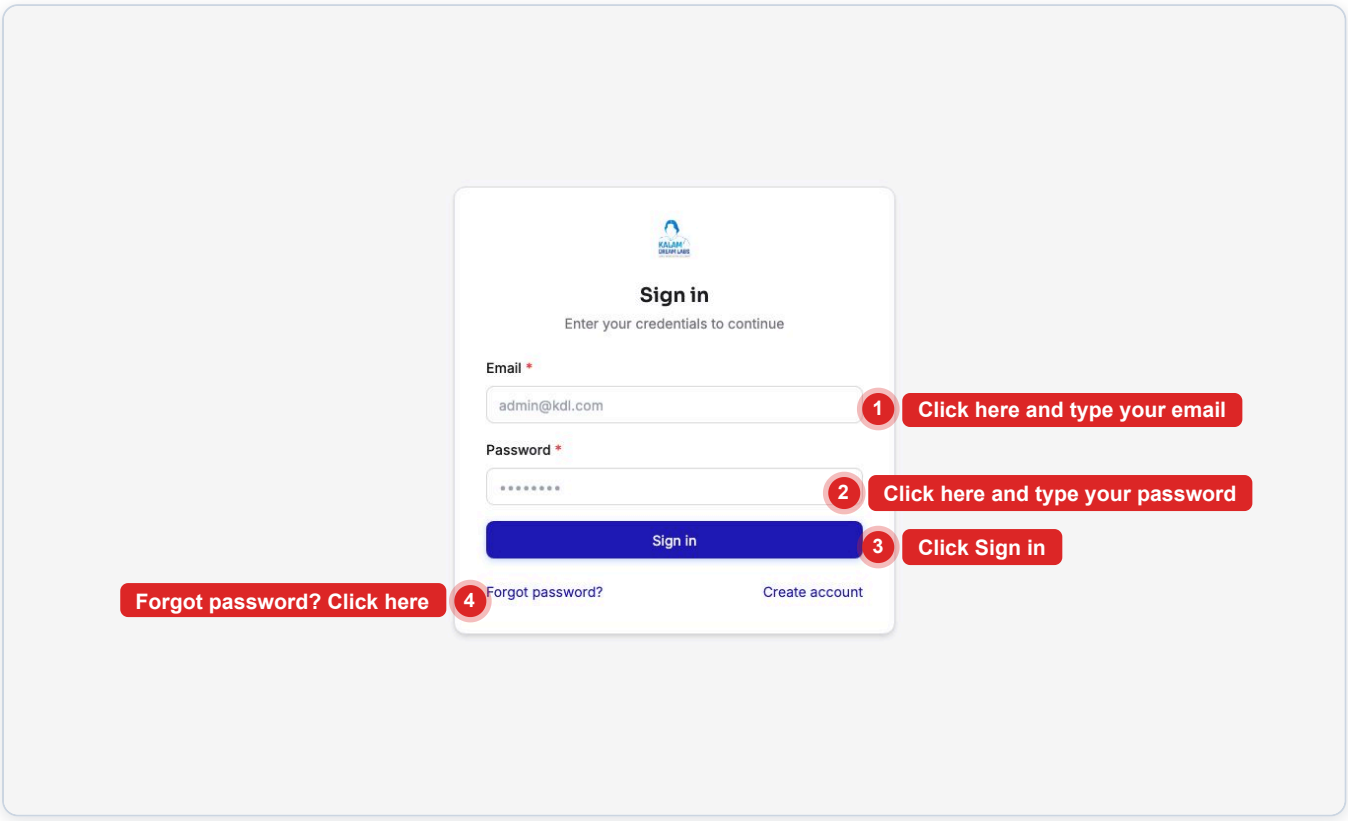
Step 5. Create the admin user

```
TERMINAL
docker compose exec backend node prisma/seed.js
```

You should see `Seeded admin: admin@kdl.com`.

Step 6. Sign in

Open **http://localhost/login**. Email `admin@kdl.com`, password `kdl-dev-seed-password`. KDL OS asks you to choose a new password on first sign-in. Then open **Admin > Documents** and follow the guide for the flow you chose.



The sign-in screen

> Forgot the password or locked out? `docker compose exec backend node scripts/reset-admin-password.js --email admin@kdl.com --password 'Admin@123'` (at least 8 characters), then sign in with it.

Notes: Everyday commands and problems

This is not a step. Run the commands inside the `kdl-os` folder.

I want to	Type
Stop (keeps data)	`docker compose down`
Start again	`docker compose up -d`
See what runs	`docker compose ps`
Update to the latest code	`git pull` then `docker compose up -d --build`
Read backend logs	`docker compose logs backend --tail 100`

> `docker compose down -v` deletes **all your data**. Do not use it unless you want to start from zero.

Problem	Do this
A variable is missing	Wrong folder, or Step 3 was skipped. Go to `kdl-os` and repeat Step 3.
Port already in use	Close the program using it (for example a local site on port 80).
Build stops with Killed or out of memory	Give Docker 8 GB (Step 1), run Step 4 again.
502 Bad Gateway	Still starting. Wait 1 to 2 minutes and refresh.
Wrong password	Did you run Step 5? The first password is `kdl-dev-seed-password`.
Account locked / 429	Wait a minute, or `docker compose restart backend`.